GTN (Glyceryl Trinitrate) IV

Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

Authorised practice levels

- CCPs.

Indications

- ✓ Cardiogenic pulmonary oedema and the patient is not rapidly improving with GTN SL and CPAP.

Contraindications

- ✗ Known severe allergy.
- ✗ Systolic BP less than 110 mmHg.
- ✗ Heart rate less than 40/minute.
- ✗ Heart rate greater than 150/minute.
- ✗ Ventricular tachycardia.

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Cautions

- STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- The patient is frail.
- Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause further fall in cardiac output.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN IV is very low, and personnel must seek clinical advice prior to administration.

Dosage

- Start the infusion at 2 mg/hour. Adjust the rate as required by 0.5 mg/hour every 5-10 minutes, titrating to effect.

Administration

- Draw up 50 mg of GTN from a 50 mg/10 ml ampoule and dilute to a total of 50 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
- Administer via an infusion device.

Common adverse effects

- Hypotension.
- Flushing.
- Headache.

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- Tachycardia.
- Feeling light-headed.

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 3-5 minutes following cessation of the infusion.

Usual preparation

- Ampoule containing 50 mg/10 ml.

Pharmacokinetics

- GTN is predominantly metabolised in the liver.

Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

Additional information

- Particular caution must be used in the presence of STEMI involving the right ventricle:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.

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- This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
- GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.

More Information

Version history

11 September 2023 1.0.4

New: Guideline released.